Clopidogrel

- Clopidogrel is a **prodrug**. It is inactive until metabolised in the liver.
- About 85% is hydrolysed by esterases (CES1) to an inactive metabolite. Only about 15% goes through the two-step CYP activation.
- Activation is by **CYP2C19** (main enzyme), with CYP3A4, CYP1A2, CYP2B6 and CYP2C9 also contributing. This produces an active thiol metabolite.
- The active metabolite **irreversibly binds the P2Y12 receptor** on platelets (via a disulfide bond).
- Blocking P2Y12 prevents ADP from lowering cAMP and from activating the **GP IIb/IIIa** receptor, so fibrinogen cannot cross-link platelets and aggregation is inhibited.
- Because binding is irreversible, the effect lasts for the **platelet lifespan (7 to 10 days)**. Platelet function recovers only as new platelets are made.

Adverse effects

- **Bleeding** (the most important), including GI bleeding
- Dyspepsia, diarrhoea, rash
- **Thrombotic thrombocytopenic purpura (TTP)**: rare but serious
- Neutropenia (rare)

Drug interactions

- **CYP2C19 inhibitors** reduce activation and lower the antiplatelet effect. The main example is **omeprazole/esomeprazole**. Pantoprazole is preferred if a PPI is needed. Others include fluoxetine, fluvoxamine and ticlopidine.
- **CYP2C19 inducers** (e.g. rifampicin) increase activation and bleeding risk.
- **NSAIDs, warfarin, aspirin** increase bleeding risk.
- **Active drug:** slow metaboliser = toxicity, rapid metaboliser = failure
- **Prodrug:** slow metaboliser = failure, rapid metaboliser = toxicity

Renal tubular secretion

Mechanism

- Many drugs are actively secreted from blood into urine by transporters on the proximal tubule cells.
    - **OAT1/OAT3** (organic anion transporters) handle acidic drugs such as penicillins.
    - **OCT2** (organic cation transporter) and **MATE1/MATE2-K** handle basic drugs.
- If two drugs use the same transporter, they **compete**. The one with the higher affinity or concentration blocks secretion of the other, so the second drug is cleared more slowly. Its plasma level, t½ and effect rise.

Probenecid and penicillin (OAT)

- Probenecid is an organic acid that competes with penicillin for **OAT1/OAT3**.
- Less penicillin is secreted, so its plasma level and half-life increase.
- **Clinical use:** deliberately combined with penicillin or ampicillin to prolong action, e.g. a single-dose regimen for gonorrhoea or neurosyphilis, or in severe infections.
- Probenecid also blocks urate reabsorption via URAT1, which is why it is used as a **uricosuric in gout**.
- Other drugs whose levels it raises: cephalosporins, methotrexate, indomethacin, dapsone, and some antivirals (e.g. aciclovir).

Cimetidine and trimethoprim (OCT/MATE)

- Cimetidine is a basic drug secreted by **OCT2 and MATE**. It competes with trimethoprim (also a base) for these cationic transporters.
- Less trimethoprim is secreted, so its level rises.
- A small caution: this effect is mainly on **trimethoprim**, not sulfamethoxazole. Sulfamethoxazole is an acid, so it is handled by the anionic pathway.
- Trimethoprim itself also inhibits creatinine secretion (via OCT2/MATE), which causes a rise in serum creatinine without a true fall in GFR. Dofetilide and metformin are other OCT/MATE substrates affected by cimetidine.

Enterohepatic circulation (EHC)

1. The drug is taken up by hepatocytes and conjugated (usually with glucuronic acid, by UGT enzymes), making it more polar.

2. The conjugate is secreted into bile (via transporters such as MRP2 and P-gp on the canalicular membrane) and stored in the gallbladder.
3. Bile enters the duodenum, usually after a meal (gallbladder contraction is stimulated by CCK).
4. Gut bacteria produce β-glucuronidase, which cleaves the glucuronide and releases the free (parent) drug.
5. The free drug is lipophilic again, so it is reabsorbed from the intestine into the portal vein and returns to the liver.
6. The cycle repeats until the drug is eventually excreted in faeces or urine.

Examples: **oral contraceptives (ethinylestradiol)**

Clinical relevance

1. Antibiotics and oral contraceptives

- Broad-spectrum antibiotics reduce gut bacteria, so less β-glucuronidase is available, and less ethinylestradiol is deconjugated and reabsorbed. Levels may fall.
- This has been proposed as a reason for contraceptive failure, but evidence is weak for most antibiotics. **Rifampicin** is the real exception, because it induces liver enzymes and clearly reduces contraceptive effect.

2. Bile acid sequestrants (cholestyramine, colestipol)

- They bind drugs and bile acids in the gut and prevent reabsorption, which breaks the cycle.
- Used to treat **overdose or toxicity** of drugs with EHC (e.g. digoxin, warfarin, leflunomide) and to lower cholesterol.

3. Activated charcoal

- Repeated doses can **interrupt EHC** and speed elimination (e.g. in overdoses of carbamazepine, phenobarbital, theophylline).

4. Mycophenolate

- Its glucuronide is recycled via EHC. Cholestyramine and some antibiotics reduce its exposure.

5. Liver or biliary disease and cholecystectomy

- Reduced bile flow means reduced EHC, which changes drug levels.